

Module 3: Configuring Interfaces and Switches

CompTIA Network+ N10-009

Module Overview

Learning Objectives

By the end of this module, you should be able to:

- Deploy networking devices.
- Explain network interfaces.
- Deploy common Ethernet switching features.
- Troubleshoot transceiver and switching issues.

Study focus: Be prepared to connect physical interface characteristics to switch configuration, performance, and troubleshooting decisions.

Lesson 3.1: Network Interfaces

Exam Objectives Covered: 1.5, 5.2

Network Interface Cards (NICs)

Definition: A hardware component that connects a computer or network device to a network and provides the interface needed to transmit and receive data.

Key Characteristics:

- Most support: copper Gigabit Ethernet.
- May provision multiple ports for:
 1. Connecting to different networks.
 2. Aggregating links to provide higher bandwidth or redundancy.
- Each NIC contains a unique: Media Access Control (MAC) address.

Important distinction: A NIC provides network connectivity, while a transceiver converts signals between the device interface and the transmission medium.

Modular Transceivers

What are modular transceivers? Removable devices that transmit and receive network signals and allow a switch or NIC to support different media types, speeds, distances, and connector types.

Modular transceivers are useful because an organization can select the appropriate copper or fiber connection without replacing the entire switch or NIC.

Types of Transceivers

1. SFP — Small Form-Factor Pluggable

- **Purpose:** Provides a removable interface for network connections.
- **Common uses:** Copper or fiber Ethernet connections at standard Gigabit Ethernet speeds.

2. SFP+ — Enhanced SFP

- **Enhancement over SFP:** Supports higher data rates while using a similar small form factor.
- **Speed capabilities:** Commonly supports up to 10 *Gbps*, depending on the module and equipment.

3. QSFP — Quad Small Form-Factor Pluggable

- **Key feature:** Uses multiple communication lanes in one transceiver.
- **Applications:** High-bandwidth data-center and backbone connections.

4. QSFP+ — Enhanced Quad Small Form-Factor Pluggable

- **Enhanced capabilities:** Provides higher performance than QSFP and commonly supports multiple high-speed lanes.
- **Typical speeds:** Commonly supports approximately 40 *Gbps*, depending on the implementation.

Exam reminder: Transceiver labels do not guarantee compatibility. Always verify the module's speed, fiber type, wavelength, connector, distance rating, and compatibility with the switch or NIC.

Transceiver Mismatch Issues

What causes transceiver mismatches?

- Incompatible fiber types.
- Mismatched transmission speeds.
- Incorrect wavelengths.
- Unsupported distances.
- Incompatible connectors or transceiver standards.
- A mismatch between duplex and parallel deployment methods.
- Using a module that the manufacturer does not support.

Fiber Type Compatibility:

- Always refer to: Manufacturer documentation.

Deployment Methods

1. **Duplex:** Uses two separate fiber strands: one for transmitting and one for receiving.
2. **Parallel:** Uses multiple fiber strands or lanes simultaneously to carry data.
3. **WDM — Wavelength Division Multiplexing:** Uses different wavelengths of light over the same fiber to transmit multiple signals.

Troubleshooting clue: If two transceivers are physically installable but the link does not come up, physical fit does not prove technical compatibility.

Transceiver Signal Strength Issues

Optical Link Budget — Loss Budget

- **Definition:** The total amount of signal loss that an optical link can tolerate while still maintaining a reliable connection.
- **Why it matters:** If the combined loss is greater than the available optical budget, the receiving device may not detect the signal correctly.

Sources of Signal Loss

1. **Attenuation:** The gradual weakening of an optical signal as it travels through the fiber.
2. **Connectors:** Connection points can introduce loss because of dirt, damage, misalignment, or poor installation.
3. **Splices:** Joined sections of fiber can introduce additional loss when the fibers are not aligned or fused correctly.

Troubleshooting steps:

- Inspect and clean fiber connectors.
- Verify the fiber type and length.
- Check the transceiver's transmit and receive power specifications.
- Inspect splices and patch panels.
- Compare measured loss with the manufacturer's link budget.

Ethernet Frame Format

An Ethernet frame packages data for delivery across a local network.

- **Preamble:** A synchronization pattern that helps the receiving device prepare to receive the frame.
- **Destination MAC Address:** Identifies the intended receiving device.
- **Source MAC Address:** Identifies the device that sent the frame.
- **EtherType Field:** Identifies the upper-layer protocol or payload type carried by the frame.
- **Data Payload:** Contains the encapsulated data being transported.
- **Frame Check Sequence — FCS:** An error-checking value used to detect corruption during transmission.

Media Access Control — MAC Address Format

Structure

- **Total length:** 48 bits (6 bytes).
- **Displayed in:** Hexadecimal notation, commonly as six groups of two hexadecimal digits, six groups separated by colons, or three groups separated by hyphens.

Example format: 00:1A:2B:3C:4D:5E

Components

OUI — Organizationally Unique Identifier

- **Length:** 6 digits (3 bytes).
- **Purpose:** Identifies the organization or manufacturer assigned to the address range.

Serial Number

- **Length:** 6 digits (3 bytes).
- **Purpose:** Identifies the individual network interface within the manufacturer's assigned address range.

Types of MAC Addresses

1. **Burned-in Address — BIA:** A MAC address assigned to a device by the manufacturer and stored in the device's hardware or firmware.
2. **Locally Administered Address:** An address manually or automatically assigned by an administrator or operating system instead of being permanently assigned by the manufacturer.
3. **Broadcast Address:** An address intended for every device on the local network. It is represented as `FF:FF:FF:FF:FF:FF`.
4. **Unicast Address:** An address intended for one specific network interface.

Lesson 3.2: Ethernet Switches

Exam Objectives Covered: 1.2, 2.2

Network Device Evolution

Hubs

Definition: A basic Layer 1 networking device that repeats incoming signals to every connected port.

How they work: A hub receives a signal on one port and sends a copy of that signal out all other ports.

Limitations:

- All devices share the same collision domain.
- Traffic is sent to devices that are not the intended recipients.
- Hubs provide little or no traffic control, security, or management capability.

Bridges

Definition: A Layer 2 device that connects network segments and forwards frames based on MAC addresses.

How they improve upon hubs: Bridges can limit unnecessary traffic by forwarding frames only toward the segment containing the destination device.

Key function: Learn MAC addresses and use the information to make forwarding decisions.

Limitations: Bridges generally support fewer ports and have less performance, scalability, and management capability than modern switches.

Switches

Definition: A multiport Layer 2 networking device that forwards Ethernet frames based on MAC addresses.

How they work: A switch learns the source MAC address of incoming frames, records the associated port in its MAC address table, and forwards frames toward the correct destination port.

Advantages over hubs and bridges:

1. Each switch port can provide a separate collision domain.
2. Frames can be forwarded selectively instead of being repeated everywhere.
3. Switches support many ports and higher performance.
4. Managed switches provide configuration, monitoring, security, VLAN, and troubleshooting features.

Ethernet Switch Types

1. Unmanaged vs. Managed Switches

- **Unmanaged:** A plug-and-play switch with little or no configuration capability.
- **Managed:** A configurable switch that can be monitored and controlled through a CLI, web interface, or network-management platform.
- **When to use each:** Use unmanaged switches for simple, low-cost networks; use managed switches when the network requires VLANs, monitoring, security, redundancy, traffic control, or detailed troubleshooting.

2. Stackable Switches

- **Definition:** Switches designed to operate together as a logical unit through special stacking connections.
- **Benefits:** Centralized management, simplified administration, increased port capacity, and improved redundancy or performance.

3. Modular vs. Fixed Switches

- **Modular:** A switch with replaceable components or expansion modules, such as additional line cards or power supplies.
- **Fixed:** A switch with a set number and type of built-in ports that generally cannot be expanded internally.

4. Desktop vs. Rack-Mounted

- **Desktop:** A compact switch designed to sit on a desk, shelf, or other flat surface.
- **Rack-Mounted:** A switch designed to be installed in a standard equipment rack, usually for server rooms and data centers.

Switch Interface Configuration

Primary Configuration Method: Command-line interface (CLI).

1. User EXEC Mode

- **Prompt symbol:** >
- **Access level:** Basic monitoring and user-level access.
- **Typical commands available:** Basic status, connectivity, and help commands.

2. Privileged EXEC Mode

- **Prompt symbol:** #
- **How to access:** Enter `enable` from User EXEC mode.
- **Additional capabilities:** Advanced monitoring, diagnostic commands, configuration viewing, and access to Global Configuration mode.

3. Global Configuration Mode

- **Prompt symbol:** (config) #
- **How to access:** Enter `configure terminal` from Privileged EXEC mode.
- **Purpose:** Configure device-wide settings and enter more specific configuration modes.

Mode sequence:

```
Switch> enable
Switch# configure terminal
Switch(config) #
```

Lesson 3.3: Switch Port Configuration

Exam Objectives Covered: 2.2, 5.3

Link Aggregation and NIC Teaming

Link Aggregation Definition: Combining multiple physical network links into one logical link.

Benefits:

- Increased total bandwidth.
- Redundancy if one physical link fails.
- Improved availability and load distribution.

NIC Teaming:

- **Purpose:** Combines multiple network interface ports on a server into one logical team for redundancy, load balancing, or increased throughput.
- **Common protocols:** Link Aggregation Control Protocol (LACP) and vendor-specific aggregation methods.

Configuration requirement: The connected devices must agree on the aggregation method and settings. A mismatch can cause suspended interfaces or loss of connectivity.

Maximum Transmission Unit — MTU

Standard Ethernet Frame

- **Maximum frame length:** 1,518 bytes.
- **Header components:** 18 bytes total:
 - Destination MAC: 6 bytes.
 - Source MAC: 6 bytes.
 - EtherType field: 2 bytes.
 - Error checking field, or FCS: 4 bytes.
- **Maximum data payload — MTU:** 1,500 bytes.

Jumbo Frames

- **Maximum size:** Up to 9,216 bytes.
- **Benefits:** Reduces the number of frames and processing overhead when transferring large amounts of data.
- **Considerations:** Every device along the communication path must support the same jumbo-frame configuration. An MTU mismatch can cause fragmentation, dropped frames, or communication failures.

Spanning Tree Protocol — STP

Purpose: Prevents Layer 2 switching loops.

Problem it solves: Redundant physical links can create loops in which frames circulate repeatedly through the network.

How it works: STP exchanges Bridge Protocol Data Units, selects a logical loop-free topology, and places redundant paths into a blocking state. If the active path fails, STP can allow a previously blocked path to forward traffic.

Spanning Tree Protocol Port States

State	Forwards Frames?	Learns MACs?	Description
Blocking	No	No	Drops all frames except BPDUs.
Listening	No	No	Listens for BPDUs and evaluates the topology for loops.
Learning	No	Yes	Learns MAC addresses and builds the MAC address table without forwarding user frames.
Forwarding	Yes	Yes	Operates normally and forwards frames.
Disabled	No	No	The port has been disabled by an administrator or is otherwise not participating.

Key Concept — BPDU: A Bridge Protocol Data Unit is a control frame used by switches to exchange STP information and identify a loop-free network topology.

Power over Ethernet — PoE

Purpose: Delivers electrical power and network data over Ethernet cabling, allowing compatible devices to operate without a separate power cable.

PoE Standards

1. 802.3af — PoE

- **Power delivery:** Provides basic power over Ethernet.
- **Typical devices:** IP phones, basic wireless access points, and simple security cameras.

2. 802.3at — PoE+

- **Power delivery:** Provides more power than 802.3af.
- **Enhanced capabilities:** Supports higher-power access points, video phones, pan-tilt-zoom cameras, and other devices with greater power requirements.

3. 802.3bt — PoE++

- **Power delivery:** Provides substantially more power by using additional wire pairs.
- **Advanced applications:** Supports high-performance wireless access points, advanced cameras, digital signage, building-management devices, and other high-power equipment.

Important consideration: The switch, cable, and powered device must support compatible PoE standards and power requirements.

Lesson 3.4: Switch Troubleshooting

Exam Objectives Covered: 5.2, 5.3, 5.5

Hardware Failure Issues

1. Power Issues

- **Symptoms:** The switch does not start, reboots unexpectedly, loses ports, or shows power-supply alarms.
- **Troubleshooting steps:** Verify the power cable, outlet, power supply, circuit, status indicators, and environmental conditions. Check whether the switch has redundant power supplies and whether both are operating.

2. Hardware Failure Issues

- **Common causes:** Failed power supplies, damaged ports, overheating, defective transceivers, faulty fans, water or physical damage, and component failure.
- **Diagnostic approach:** Inspect the device, check logs and status indicators, test with known-good cables and transceivers, move the connection to another port, and determine whether the problem follows the component or remains with the port.

3. Data Link Layer Issues

- **Indicators:** Intermittent connectivity, excessive frame errors, a port that is blocked or suspended, MAC-address instability, or a link that is up but unable to pass traffic correctly.
- **Resolution methods:** Verify speed and duplex settings, check VLAN and aggregation configuration, inspect the MAC address table, check for loops, replace faulty components, and review interface counters.

Port Status Indicators — LED Meanings

LED Status	Meaning
Solid green	The link is connected, but there is no current traffic.
Flickering green	The link is operating normally with traffic; the blink rate may indicate link speed.
No light	The link is not working, the cable is disconnected, or the port is shut down.
Blinking amber	A fault has been detected.
Solid amber	The port is blocked by STP to help prevent a switching loop.

Note: LED meanings can vary by manufacturer. Use the device documentation to confirm the exact meaning.

Switch Show Commands

Privileged Mode Commands

- **show config:** Displays the device configuration or configuration information.
- **show interface:** Displays interface status, counters, errors, speed, duplex, and other operational details.

Interface Status Diagnosis

1. Down/down

- **Layer 1 status:** Down.
- **Layer 2 status:** Down.
- **Meaning:** The physical link is not detected, often because of a disconnected or faulty cable, failed transceiver, inactive device, or incorrect physical connection.

2. Administratively down/down

- **Meaning:** The interface has been intentionally disabled in configuration. The port may need to be enabled by an administrator.

3. Down/error disabled

- **Meaning:** The switch has disabled the port because it detected a serious violation or fault.
- **Common causes:** Port-security violations, excessive errors, BPDU-related protection, or other protective features.

4. Up/down — Suspended

- **Meaning:** The physical link is active, but the interface is not forwarding traffic because of a configuration problem.
- **Common causes:** Link aggregation mismatch, incorrect negotiation settings, VLAN issues, or an inconsistent port configuration.

Interface Error Counters

Why monitor error counters? Increasing counters indicate that a physical, data-link, configuration, or compatibility problem may be affecting reliability.

1. CRC — Cyclic Redundancy Check Errors

- **What they indicate:** Frames arrived with corrupted data and failed the FCS verification.
- **Possible causes:** Damaged cables, electromagnetic interference, poor connectors, faulty transceivers, mismatched speed or duplex settings, or failing hardware.

2. Runt Frame Errors

- **Definition:** Frames smaller than the minimum valid Ethernet frame size.
- **Common causes:** Collisions, duplex mismatches, faulty network devices, or damaged cabling.

3. Giant Frame Errors

- **Definition:** Frames larger than the configured maximum frame size.
- **Common causes:** MTU mismatch, incorrect jumbo-frame configuration, faulty equipment, or frames arriving from a device using an unsupported size.

MAC Address Table

Purpose: Maps learned source MAC addresses to switch ports so the switch can forward frames efficiently.

How switches learn MAC addresses: When a frame enters a port, the switch examines the source MAC address and records that address as reachable through the incoming port.

CAM — Content Addressable Memory

- **Definition:** High-speed memory used by switches to store MAC address and port mappings.
- **Function:** Allows the switch to look up a destination MAC address quickly and select the appropriate outgoing port.

Flooding

- **When it occurs:** When the destination MAC address is unknown, or when the frame is a broadcast.
- **How it works:** The switch sends the frame out all appropriate ports except the port on which the frame arrived.

Security and performance concern: Excessive flooding can consume bandwidth and may indicate an unknown destination, a recently cleared MAC table, a network loop, or a broadcast storm.

Network Loop and Broadcast Storm Issues

Network Loop

- **Definition:** A Layer 2 path in which frames can circulate continuously between switches.
- **How it forms:** Redundant links are connected without proper STP operation or another loop-prevention mechanism.
- **Consequences:** Duplicate frames, unstable MAC address tables, excessive bandwidth use, broadcast storms, and widespread network unavailability.

Broadcast Storm

- **Definition:** An overwhelming volume of broadcast traffic that consumes network bandwidth and device-processing capacity.
- **Impact on network:** Devices become slow or unreachable, switch resources are exhausted, and legitimate traffic may be delayed or dropped.
- **Symptoms:** High interface utilization, rapidly increasing broadcast counts, unstable connectivity, excessive CPU use, and widespread packet loss.

How STP Prevents Loops

STP identifies redundant paths and places selected ports into a blocking state. Only the best available paths forward traffic. If an active path fails, STP can transition a redundant path toward forwarding.

Power over Ethernet Troubleshooting

1. Cabling Issues

- **Requirements:** Compatible Ethernet cabling with the required wire pairs, proper termination, acceptable length, and sufficient quality.
- **Common problems:** Damaged cables, incorrect pinouts, poor terminations, excessive cable length, unsupported cable category, and insufficient pairs for the required PoE standard.

2. Incorrect Standard

- **Problem:** The switch may provide a lower PoE class than the powered device requires, or the device may not support the standard supplied by the switch.
- **Solution:** Verify compatibility among the switch, cable, and powered device. Use a compatible PoE port, injector, or external power source.

3. Power Budget Exceeded

- **Definition:** The total power requested by connected devices is greater than the switch's available PoE power.
- **How to resolve:** Disconnect or relocate devices, prioritize critical ports, use a switch with a larger power budget, add a PoE switch or injector, or reduce the power requirements of connected devices.

Module Summary & Key Takeaways

Critical Configuration Concepts

1. Transceivers must match the required speed, fiber type, wavelength, connector, distance, and deployment method.
2. Switch ports must be configured consistently for speed, duplex, VLAN, aggregation, MTU, and security features.
3. STP, MAC address learning, and PoE must be monitored to maintain availability and prevent switching problems.

Network Interface Best Practices

- **Speed configuration:** Ensure both ends support and negotiate the same speed or are manually configured consistently.
- **Duplex settings:** Use matching duplex settings; a speed or duplex mismatch can cause collisions and errors.
- **Protocol settings:** Verify compatible Ethernet, link aggregation, VLAN, STP, MTU, and PoE settings.

Essential Switching Features

- **STP benefits:** Prevents Layer 2 loops and reduces the risk of broadcast storms.
- **Port security purpose:** Restricts which devices or MAC addresses can use a port and helps prevent unauthorized access.
- **Network segmentation:** Separates traffic into logical sections, improving performance, security, and manageability.

Troubleshooting Approach

1. **Check transceiver compatibility:** Verify media type, speed, wavelength, connector, distance, and manufacturer support.
2. **Verify configuration:** Check speed, duplex, VLAN, MTU, aggregation, STP, and PoE settings.
3. **Use diagnostic commands:** Review configuration, interface status, logs, MAC tables, and interface counters.
4. **Monitor error counters:** Look for CRC, runt, giant, collision, and other counters that increase over time.

Vocabulary Terms

- **NIC — Network Interface Card:** Hardware that connects a device to a network.
- **MAC Address:** A Layer 2 address associated with a network interface.
- **Transceiver:** A device that transmits and receives network signals.
- **SFP/SFP+/QSFP:** Families of modular network transceivers.
- **WDM — Wavelength Division Multiplexing:** A method of transmitting multiple optical signals using different wavelengths on one fiber.
- **MTU — Maximum Transmission Unit:** The largest data payload that can be carried in one frame without fragmentation.
- **Jumbo Frame:** An Ethernet frame larger than the standard maximum frame size.
- **Link Aggregation:** Combining multiple physical links into one logical connection.
- **STP — Spanning Tree Protocol:** A protocol that prevents Layer 2 switching loops.
- **BPDU — Bridge Protocol Data Unit:** A control frame used by STP-enabled switches.
- **PoE — Power over Ethernet:** Delivery of power and data over Ethernet cabling.
- **CAM — Content Addressable Memory:** High-speed memory used to store MAC address mappings.
- **Broadcast Storm:** Excessive broadcast traffic caused by a loop or other network problem.
- **CRC — Cyclic Redundancy Check:** An error-detection method used to identify corrupted frames.
- **CLI — Command-Line Interface:** A text-based interface used to configure and manage network devices.